Efficient CSP Algorithm With Spatio-Temporal Filtering for Motor Imagery Classification

Aimin Jiang, *Member, IEEE*, Jing Shang, Xiaofeng Liu, Yibin Tang, Hon Keung Kwan, *Senior Member, IEEE*, and Yanping Zhu

***Abstract*—Common spatial pattern (CSP) is an efficient algorithm widely used in feature extraction of EEG-based motor imagery classification. Traditional CSP depends only on spatial filtering, that aims to maximize or minimize the ratio of variances of filtered EEG signals in different classes. Recent advances of CSP approaches show that temporal filtering is also preferable to extract discriminative features. In view of this perspective, a novel spatio-temporal filtering strategy is proposed in this paper. To improve computational efficiency and alleviate the overfitting issue frequently encountered in the case of small sample size, the same temporal filter is designed by EEG signals of the same class and shared by all the spatial channels. Spatial and temporal filters can be updated alternatively in practice. Furthermore, each of the resulting designs can still be cast as a CSP problem and tackled efficiently by the eigenvalue decomposition. To alleviate the adverse effects of outliers or noisy EEG channels, sparse spatial or temporal filters can also be achieved by incorporating an ℓ_1-norm-based regularization term in our CSP problem. The regularized spatial or temporal filter design is iteratively reformulated as a CSP problem via the reweighting technique. Two sets of motor imagery EEG data of BCI competitions are used in our experiments to verify the effectiveness of the proposed algorithm.**

***Index Terms*—Alternative optimization scheme, brain–computer interface (BCI), common spatial pattern (CSP), electroencephalograph (EEG), generalized eigenvalue decomposition, ℓ_1 norm, motor imagery, reweighting technique, spatio-temporal filters, sparsity.**

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I. Introduction

A BRAIN-COMPUTER interface (BCI) system aims to measure and decode brain signals, such that users with severe motor disabilities can control a device through a non-muscular way [1]–[5]. In various BCI applications, brain activities are usually measured by electroencephalograph (EEG), due to its high portability and low cost. However, EEG signals are vulnerable to noise and interferences. Also, they are essentially nonstationary, which indicates that their characteristics evolve over time. Hence, appropriate signal processing and feature extraction are important to the success of EEG-based BCIs.

In this paper, we focus on motor imagery (MI) tasks, whose objective is to discriminate different brain states during visual and motor imagery [6]–[9]. Common spatial pattern (CSP) [10]–[12] is widely used in EEG-based MI BCIs. It designs subject-dependent spatial filters, which linearly combine different channels to maximize the variance of the filtered signals in one class and minimize it in the other class. A variety of extensions have been reported in the literature to enhance its performance. One class of approaches exploit simultaneous spatial and temporal filtering [13]–[15] to find more discriminative features. In [13] and [14], an augmented data matrix is constructed by the original EEG signals and their temporally-delayed copies. The usual CSP technique is then applied to obtain a spatial filter and a set of channel-sensitive temporal filters. However, the risk of overfitting is accordingly increased because more modeling parameters are introduced in [13] and [14]. Applying a common finite impulse response (FIR) filter on EEG signals of each channel, a more general CSP problem using joint spatio-temporal filtering is considered in [15]. But since the problem formulation is based on the convolution, its optimization depends on sophisticated numerical techniques and computational efficiency is damaged.

To enhance the generality, regularizations are widely used by many CSP-based approaches [14]–[18]. The ℓ_1 norm and its variants (e.g., ℓ_1/ℓ_2 norm) are the most favorable one used in practice. Applying it on a spatial filter, EEG channels with intensive noise and interferences could be identified and removed. If it is applied on a temporal filter, the computational complexity of filtering operation can be reduced by ignoring arithmetic operations corresponding to zero filter coefficients. Moreover, the sparsity is also useful to alleviate the overfitting especially when dealing with a small amount of samples. The aforementioned approaches regularizes a specific CSP problem by incorporating a sparsity-encouraging term in its objective function. The sparsity of filter coefficients can also be directly minimized subject to an appropriate performance constraint [19] and [20].

In [21] and [22], the Riemannian manifold is used to model the positive-definite covariance structure of EEG data, such that filtering and classification procedures can be merged together. In [21], using the Riemannian geodesic distance, classification can be implemented directly on the covariance matrix of an unknown trial. A covariance matrix can also be projected to its Riemannian tangent space, such that traditional classification approaches can be applied on feature vectors in the Riemannian tangent space [21]. To alleviate overfitting and avoid heavy computation in a high-dimensional space, a sub-manifold learning algorithm is derived in [22], where an intrinsic mapping matrix is learned to map a high-dimensional covariance matrix into a low-dimensional space. With the recent advances of deep learning, various deep neural networks have also been employed to deal with the classification of EEG signals [23]–[27]. Compared to traditional CSP and its variants, both manifold-based and deep-network-based approaches need more training data to achieve good performance. Unfortunately, this is not always feasible in practical applications. Furthermore, due to the low signal-to-noise ratio (SNR) and essential nonstationarity of EEG signals, suitable signal processing techniques still have to be deployed in those approaches.

In this paper, we still focus on the CSP technique with joint spatio-temporal filtering. To improve computational efficiency and avoid overfitting, in our algorithm the same temporal filter is shared by each channel of EEG signals in the same class. In practice, spatial and temporal filters are alternatively updated, such that the resulting problems can be tackled by usual CSP technique. Furthermore, an ℓ_1-norm regularization term is introduced into our filter design problems. To enhance computational efficiency, the reweighting technique is employed by the proposed algorithm, such that the resulting problem can still be formulated in the classical CSP form and efficiently solved by the generalized eigenvalue decomposition (EVD).

The paper is organized as follows. Some related work is first reviewed in Section II. An efficient spatial and temporal filter design algorithm is then developed in Section III. The sparsity-encouraging regularization is also considered in Section III. Experimental results are presented in Section IV. Section V concludes the paper.

II. Related Work

A. Traditional CSP and Sparse CSP

Let $\mathbf{X}_i = [\mathbf{x}_{i,1}\ \mathbf{x}_{i,2}\ \ldots\ \mathbf{x}_{i,K}] \in \mathbb{R}^{M\times K}$ denote a data block of the ith trial, where M and K represent, respectively, the numbers of channels and sampling points in the data block. The traditional CSP problem is formulated as

$$\max_{\mathbf{w}\in\mathbb{R}^M} \frac{\mathbf{w}^T\mathbf{C}_1\mathbf{w}}{\mathbf{w}^T\mathbf{C}_2\mathbf{w}} \tag{1}$$

where $\mathbf{w}$ represents the coefficient vector of a spatial filter, and $\mathbf{C}_k$ $(k = 1, 2)$ denotes the sample covariance matrix of EEG signals in one class, i.e.,

$$\mathbf{C}_k = \frac{1}{|\mathcal{C}_k|}\sum_{i\in\mathcal{C}_k}\mathbf{X}_i\mathbf{X}_i^T, \quad k = 1, 2. \tag{2}$$

Here, $\mathcal{C}_k$ denotes the index set of data blocks associated with the kth mental state. It is known that the optimal solution to (1) is obtained by simultaneous diagonalization of two covariance matrices. More specifically, let the generalized EVD of $\mathbf{C}_1$ and $\mathbf{C}_1+\mathbf{C}_2$ be

$$\mathbf{C}_1\mathbf{u}_j = \lambda_j\left(\mathbf{C}_1+\mathbf{C}_2\right)\mathbf{u}_j, \tag{3}$$

where λ_j denotes a generalized eigenvalue and $\mathbf{u}_j$ is the corresponding eigenvector. Without the loss of generality, eigenvalues $\{\lambda_j\}$ are arranged in a non-ascending order, i.e., $\lambda_1 \ge \lambda_2 \ge \ldots \ge \lambda_M$. Then, the optimal solution to (1) is given by the leading eigenvector $\mathbf{u}_1$.

Since EEG signals are vulnerable to noise and interferences, some researchers aim to improve the robustness of CSP algorithm by forcing some spatial coefficients equal to zeros, such that the corresponding noisy channels can be removed. For example, the regularized CSP algorithm proposed in [17] introduces the coefficient sparsity by ℓ_1/ℓ_2 norm, yielding the following optimization problem

$$\min_{\{\mathbf{w}_i\}} f(\mathbf{w}_1,\ldots,\mathbf{w}_{2N}) \tag{4a}$$

$$\text{s.t. } \mathbf{w}_i^T\left(\mathbf{C}_1+\mathbf{C}_2\right)\mathbf{w}_i = 1, \quad i = 1, 2, \ldots, 2N \tag{4b}$$

$$\mathbf{w}_i^T\left(\mathbf{C}_1+\mathbf{C}_2\right)\mathbf{w}_j = 0, \quad i \neq j \tag{4c}$$

where N denotes the number of spatial filters used for feature extraction of one class, and the objective function is defined by

$$f(\mathbf{w}_1,\ldots,\mathbf{w}_{2N}) = (1-\rho)\sum_{i=1}^{2N}\frac{\|\mathbf{w}_i\|_1}{\|\mathbf{w}_i\|_2} + \rho\left(\sum_{i=1}^{N}\mathbf{w}_i^T\mathbf{C}_2\mathbf{w}_i + \sum_{i=N+1}^{2N}\mathbf{w}_i^T\mathbf{C}_1\mathbf{w}_i\right). \tag{5}$$

The above problem is nonconvex and, thus, there is no guarantee that the optimal solution to (4) is definitely achieved.

In practice, maximizing $\frac{\mathbf{w}^T\mathbf{C}_1\mathbf{w}}{\mathbf{w}^T\mathbf{C}_2\mathbf{w}}$ does not always lead to performance improvement. Thereby, the sparse CSP problem considered in [19] is to maximize the sparsity of spatial filters, when the ratio is above some appropriate threshold. The resulting problem is formulated as

$$\min_{\mathbf{w}\in\mathbb{R}^M} \|\mathbf{w}\|_1 \tag{6a}$$

$$\text{s.t. } \frac{\mathbf{w}^T\mathbf{C}_1\mathbf{w}}{\mathbf{w}^T\mathbf{C}_2\mathbf{w}} \ge \tau \tag{6b}$$

$$\|\mathbf{w}\|_2 = 1 \tag{6c}$$

where τ is a predefined threshold used to control the tradeoff between classification accuracy and sparsity. The above problem can be tackled by the convex relaxation technique [19] or the iterative reweighting approach [20]. Experimental results show that the above algorithm can identify and discard a significant number of EEG channels with a limited degradation of classification accuracy.

B. CSP With Spatio-Temporal Filtering

Traditional CSP aforementioned depends on spatial filtering. Recent advances show that its classification performance can be further improved by adopting simultaneous filtering in both spatial and temporal domains [14] and [15].

A spatio-temporally regularized CSP (STRCSP) algorithm is developed in [14]. Let $\mathbf{X}_i^{(\tau)} \in \mathbb{R}^{M\times K}$ denote an EEG data block of the ith trial delayed by τ time steps. Then, an augmented data matrix is further defined by

$$\tilde{\mathbf{X}}_i = \begin{bmatrix} \mathbf{X}_i^{(0)} \\ \mathbf{X}_i^{(1)} \\ \vdots \\ \mathbf{X}_i^{(L-1)} \end{bmatrix}. \tag{7}$$

The traditional CSP approach is implemented on $\left\{\tilde{\mathbf{X}}_i\right\}$ to obtain a spatio-temporal filter $\mathbf{w}$ of size ML, where L represents the length of temporal filters. Assuming that these temporal filters correspond to different spatial channels, that is, $\mathbf{h}_c = [h_{c1}\ h_{c2}\ \ldots\ h_{cL}]^T$ for $c = 1,\ldots,M$, $\mathbf{w}$ can be expressed as $\mathbf{w} = \left[\mathbf{w}_1^T\ \mathbf{w}_2^T \ldots \mathbf{w}_L^T\right]^T$ where $\mathbf{w}_l = [s_1 h_{1l}\ \ldots\ s_M h_{Ml}]^T$. Here, $\mathbf{s} = [s_1\ s_2\ \ldots\ s_M]^T$ could be considered as a pure spatial filter. Although using different $\mathbf{h}_c$ for each channel leads to a tractable CSP problem, given $\mathbf{w}$, $\mathbf{s}$ and $\{\mathbf{h}_c\}$ are not uniquely determined if no assumptions are further introduced. Moreover, ML coefficients are used to extract classification features for EEG signals of each class, which increases the overfitting risk of the STRCSP algorithm. As an attempt to avoid this issue, an l_2-norm regularization term is incorporated in the denominator of the CSP problem (1) based on the augmented data matrix $\left\{\tilde{\mathbf{X}}_i\right\}$.

C. Classification Algorithms Based on Riemannian Geometry

In both traditional and spatio-temporal CSP algorithms, the obtained filters are used to extract low-dimensional features, that are fed to specific classifiers, such as linear discriminant analysis (LDA) [28], [31] and support vector machine (SVM) [28], [32]. Another group of algorithms work directly on covariance matrices $\{\mathbf{C}_k\}$, using the concept of Riemannian geometry in the manifold of covariance matrices [21] and [22].

Sample covariance matrix $\mathbf{P}_i$ of the ith trial is symmetric and positive definite (SPD). The space $\mathcal{SPN}(M)$ of SPD matrices endowed with Riemannian distance defined below is a differentiable Riemannian manifold

$$\delta_R\left(\mathbf{P}_i,\mathbf{P}_j\right) = \left\|\log\left(\mathbf{P}_i^{-1}\mathbf{P}_j\right)\right\|_F, \quad \forall \mathbf{P}_i,\ \mathbf{P}_j \in \mathcal{SPN}(M). \tag{8}$$

Based on the above metric, one can obtain the Riemannian mean of covariance matrices $\{\mathbf{P}_i\}_{i\in\mathcal{C}_k}$ by solving

$$\bar{\mathbf{P}}_k = \underset{\mathbf{P}\in\mathcal{SPN}(M)}{\arg\min} \sum_{i\in\mathcal{C}_k} \delta_R^2\left(\mathbf{P},\mathbf{P}_i\right). \tag{9}$$

Then, the classification of new EEG data is achieved by measuring the distances of their covariance matrices to the Riemannian means of different classes.

Given the Riemannian mean $\bar{\mathbf{P}}$ of the whole dataset, each sample covariance matrix $\mathbf{P}_i$ can also be mapped into the tangent space, yielding a tangent vector $\mathbf{p}_i$

$$\mathbf{p}_i = \text{upper}\left(\log\left(\bar{\mathbf{P}}^{-\frac{1}{2}}\mathbf{P}_i\bar{\mathbf{P}}^{-\frac{1}{2}}\right)\right) \tag{10}$$

where upper$(\cdot)$ operator extracts upper triangular matrix in a vector form. Once $\{\mathbf{p}_i\}$ are obtained, any kind of classification algorithms (e.g., LDA and SVM) can be implemented directly.

To alleviate the risk of overfitting, a bilinear mapping $\mathbf{S}_i = \mathbf{W}_s\mathbf{P}_i\mathbf{W}_s^T$, where $\mathbf{W}_s \in \mathbb{R}^{\tilde{M}\times M}$ and $\tilde{M} < M$, is learned in [22] by solving

$$\mathbf{W}_{opt} = \underset{\mathbf{W}_s}{\arg\min} \sum_{i,j} \left|\delta_R\left(\mathbf{P}_i,\mathbf{P}_j\right) - \delta_R\left(\mathbf{S}_i,\mathbf{S}_j\right)\right|. \tag{11}$$

Then, the classification can be further implemented on $\mathbf{S}_i$ with a reduced dimension.

III. Proposed Spatio-Temporal Filter Design

A. Spatio-Temporal Filter Designs

In traditional CSP, only spatial filters are employed to extract features. In essence, EEG signals are highly nonstationary. To improve classification accuracy, temporal filters can also be applied on EEG signals of each channels. To this end, we shall develop in this subsection an efficient algorithm to design spatial and temporal filters both based on the CSP technique.

Given data matrices $\{\mathbf{X}_i\}$, the joint spatio-temporal filter design problem can be formulated as

$$\max_{\mathbf{w}\in\mathbb{R}^M,\mathbf{h}\in\mathbb{R}^L} \frac{\mathbf{w}^T\mathbf{C}_1^h\mathbf{w}}{\mathbf{w}^T\mathbf{C}_2^h\mathbf{w}} \tag{12}$$

where $\mathbf{h} = [h_1\ \ldots\ h_L]^T$ is the coefficient vector of a temporal filter and $\left\{\mathbf{C}_k^h\right\}_{k=1}^2$ denote sample covariance matrices of EEG signals after temporal filtering by $\mathbf{h}$. In view of (2), $\mathbf{C}_k^h$ is computed by

$$\mathbf{C}_k^h = \frac{1}{|\mathcal{C}_k|}\sum_{i\in\mathcal{C}_k}\mathbf{X}_i\mathbf{H}^T\mathbf{H}\mathbf{X}_i^T, \quad k = 1,2 \tag{13}$$

where

$$\mathbf{H} = \begin{bmatrix} h_1 & \ldots & h_L & & & \\ & h_1 & \ldots & h_L & & \\ & & \ddots & \ddots & \ddots & \\ & & & h_1 & \ldots & h_L \end{bmatrix}. \tag{14}$$

Using temporally-delayed copies of EEG signals to construct an augmented data matrix $\tilde{\mathbf{X}}_i$ as in (7), (12) can also be formulated as

$$\max_{\tilde{\mathbf{w}}\in\mathbb{R}^{ML}} \frac{\tilde{\mathbf{w}}^T\tilde{\mathbf{C}}_1\tilde{\mathbf{w}}}{\tilde{\mathbf{w}}^T\tilde{\mathbf{C}}_2\tilde{\mathbf{w}}} \tag{15a}$$

$$\text{s.t. } \tilde{\mathbf{w}} = \mathbf{w}\otimes\mathbf{h}, \tag{15b}$$

where $\otimes$ represents the Kronecker product, and $\tilde{\mathbf{C}}_k$ denotes the sample covariance matrix of $\left\{\tilde{\mathbf{X}}_i\right\}_{i\in\mathcal{C}_k}$. If (15b) is removed,

Algorithm 1 Alternating Optimization Algorithm for Joint Spatio-Temporal Filter Designs (JSTFD).

Set $l = 0$ and initialize $\mathbf{h}^{(0)}$ as 1 or $\mathbf{H}^{(0)}$ as $\mathbf{I}_K$;
while not converged **do**
  Set $l = l + 1$;
  Given $\mathbf{h}^{(l-1)}$, construct $\mathbf{C}_k^h$ for $k = 1, 2$ using (13);
  Solve (12) for $\mathbf{w}^{(l)}$;
  Given $\mathbf{w}^{(l)}$, construct $\mathbf{C}_k^w$ for $k = 1, 2$ using (17);
  Solve (16) for $\mathbf{h}^{(l)}$.
end while

the above problem is reduced to the formulation of spatio-temporal filtering considered in [13] and [14], which can be tackled by usual CSP technique. However, without (15b) the number of tunable coefficients becomes ML. Compared to our model, this increases the overfitting risk and could impair the generality of the whole classification system. On the other hand, sharing $\mathbf{h}$ among spatial channels of EEG signals in the same class yields the constrained CSP problem (15). But the existence of (15b) essentially increases the optimization difficulty of problem (15), which cannot be simply tackled by the generalized EVD of $\tilde{\mathbf{C}}_1$ and $\tilde{\mathbf{C}}_1 + \tilde{\mathbf{C}}_2$.

B. Alternating Optimization Algorithm

Jointly optimizing spatial and temporal filters in (12) is still a challenging task. In practice, we adopt an alternating optimization algorithm to solve (12), which means that $\mathbf{w}$ and $\mathbf{h}$ are updated alternatively. Furthermore, it will be shown that each optimization problem can be formulated in the CSP framework and efficiently tackled by the generalized EVD.

At each iterative step, we fix either spatial or temporal filter and solve for the other one. This procedure continues until none of filter coefficients are significantly updated or the iteration number l is larger than a specific value T_1 (e.g., $T_1 = 20$ in our experiments). The major steps of the alternating optimization algorithm are summarized in Algorithm 1 and more details are described below.

Given $\mathbf{h}^{(l-1)}$ obtained in the previous iteration, (12) is solved by the generalized EVD to obtain $\mathbf{w}^{(l)}$. When updating the temporal filter, (12) is reformulated as

$$\max_{\mathbf{h}\in\mathbb{R}^L} \frac{\mathbf{h}^T\mathbf{C}_1^w\mathbf{h}}{\mathbf{h}^T\mathbf{C}_2^w\mathbf{h}} \tag{16}$$

where

$$\mathbf{C}_k^w = \frac{1}{|\mathcal{C}_k|}\sum_{i\in\mathcal{C}_k}\mathbf{W}^{(l)}\hat{\mathbf{X}}_i\hat{\mathbf{X}}_i^T\mathbf{W}^{(l)T}, \quad k = 1, 2 \tag{17}$$

$$\mathbf{W}^{(l)} = \mathbf{I}_L \otimes \mathbf{w}^{(l)T} \tag{18}$$

$$\hat{\mathbf{X}}_i = \begin{bmatrix} \mathbf{x}_{i,1} & \mathbf{x}_{i,2} & \cdots & \mathbf{x}_{i,K-L+1} \\ \mathbf{x}_{i,2} & \mathbf{x}_{i,3} & \cdots & \mathbf{x}_{i,K-L+2} \\ \vdots & \vdots & \cdots & \vdots \\ \mathbf{x}_{i,L} & \mathbf{x}_{i,L+1} & \cdots & \mathbf{x}_{i,K} \end{bmatrix} \tag{19}$$

where $\mathbf{I}_L$ represents an $L \times L$ identity matrix. Obviously, (16) can still be efficiently solved by the generalized EVD.

The computational complexity of the proposed algorithm in each iterative step is dominated by matrix multiplication and the generalized EVD. Note that, whenever computing $\mathbf{C}_k^h$ or $\mathbf{C}_k^w$, the practical computation of matrix multiplication can be significantly reduced, since coefficient matrices $\mathbf{H}$ in (14) and $\mathbf{W}$ in (18) are both sparse. So the computational complexity of Algorithm 1 is $\mathcal{O}\left(M^3 + L^3\right)$ per iteration.

If multiple spatial filters (namely, $\mathbf{w}_1$, $\mathbf{w}_2$, ..., $\mathbf{w}_N$) are required in the stage of feature extraction, the same number of temporal filters can also be designed by the proposed algorithm, each corresponding to one spatial filter. This means that the number of spatial and temporal filter coefficients to be optimized increases to $2N(M + L)$. But, when dealing with a small volume of data samples, it could lead to the overfitting issue. To address this problem, we only design one common temporal filter for each class of EEG signals in the same class, even though multiple spatial filters $\{\mathbf{w}_i\}_{i=1}^N$ are employed. So the overall number of filter coefficients is reduced to $2(NM + L)$. In this case, (17) has to be modified to

$$\mathbf{C}_k^w = \frac{1}{N\cdot|\mathcal{C}_k|}\sum_{i\in\mathcal{C}_k}\sum_{j=1}^{N}\mathbf{W}_j^{(l)}\hat{\mathbf{X}}_i\hat{\mathbf{X}}_i^T\mathbf{W}_j^{(l)T}, \tag{20}$$

where $\mathbf{W}_j^{(l)}$ for $j = 1, \ldots, N$ are obtained by taking $\mathbf{w}_j^{(l)}$ into (18).

C. Sparse Spatio-Temporal Filter Designs

The purposes of adopting sparse spatio-temporal filters are twofold.

1) EEG signals are highly noisy and contaminated with artifacts. The sparsity regularization can significantly improve the robustness of classification methods. For instance, when optimizing spatial filter $\mathbf{w}$, ℓ_1 norm can be used to identify and remove in subsequent operations spatial channels with EEG signals contaminated by severe noise and interferences.
2) The sparsity also reduces the implementation complexity of spatial and temporal filters in practice [33]–[35]. Arithmetic operations or even circuit components corresponding to zero coefficients can be eliminated or deactivated.

To achieve sparse filters, an ℓ_1-norm regularization term is further incorporated in both (12) and (16). Note that, given $\mathbf{h}$, we always have

$$\arg\max_{\mathbf{w}\in\mathbb{R}^M} \frac{\mathbf{w}^T\mathbf{C}_1^h\mathbf{w}}{\mathbf{w}^T\mathbf{C}_\Sigma^h\mathbf{w}} = \arg\max_{\mathbf{w}\in\mathbb{R}^M} \frac{\mathbf{w}^T\mathbf{C}_1^h\mathbf{w}}{\mathbf{w}^T\mathbf{C}_2^h\mathbf{w}}, \tag{21}$$

where

$$\mathbf{C}_\Sigma^h = \mathbf{C}_1^h + \mathbf{C}_2^h. \tag{22}$$

So we modify (12) to

$$\max_{\mathbf{w}\in\mathbb{R}^M} \frac{\alpha\mathbf{w}^T\mathbf{C}_1^h\mathbf{w} - (1-\alpha)\|\mathbf{w}\|_1}{\mathbf{w}^T\mathbf{C}_\Sigma^h\mathbf{w}}. \tag{23}$$

where α is a regularization parameter used to control the relative importance of the sparsity of $\mathbf{w}$.

To tackle the above problem, we reformulate $\|\mathbf{w}\|_1$ as $\|\mathbf{w}\|_1 = \mathbf{w}^T \mathbf{\Omega} \mathbf{w}$ where $\mathbf{\Omega}$ is a diagonal matrix whose ith diagonal entry is defined by $\mathbf{\Omega}_{i,i} = \frac{1}{|w_i|}$. In practice, $\mathbf{w}$ are unknown variables and, thus, $\mathbf{\Omega}$ cannot be computed beforehand. An iterative procedure nested in the alternating optimization algorithm described in the previous subsection is further introduced to update spatial filter coefficients. At each inner iteration, (23) is cast as

$$\mathbf{w}^{(l,s)} = \arg\max_{\mathbf{w}\in\mathbb{R}^M} \frac{\mathbf{w}^T \left[\alpha \mathbf{C}_1^h - (1-\alpha)\,\mathbf{\Omega}^{(l,s-1)}\right]\mathbf{w}}{\mathbf{w}^T \mathbf{C}_\Sigma^h \mathbf{w}}, \tag{24}$$

where the diagonal element of $\mathbf{\Omega}^{(l,s-1)}$ is computed by

$$\mathbf{\Omega}_{i,i}^{(l,s-1)} = \left(\eta + \left|w_i^{(l,s-1)}\right|\right)^{-p}. \tag{25}$$

Here, $w_i^{(l,s-1)}$ is the ith spatial filter coefficient obtained in the previous inner iteration. Positive constant η (e.g., 10^{-8} used in our experiments) is introduced to avoid division by zero, and $p \geq 1$ is used to accelerate the convergence. Here, we use the superscript $^{(l,s)}$ to indicate that (24) is nested in the lth iterative step of Algorithm 1. Problem (24) can still be viewed as a CSP problem and efficiently solved by the generalized EVD. Its solution is taken into (25) to update $\mathbf{\Omega}^{(l,s)}$. The inner iterative procedure is terminated when the update of $\mathbf{w}^{(l,s)}$ is negligible or the specified maximum iteration number T_2 (e.g., $T_2 = 500$) is reached. Once the inner iterative procedure converges, the final solution is returned as $\mathbf{w}^{(l)}$ to the outer iterative procedure.

The same iterative reweighting algorithm described above is also employed to sparsify temporal filters. Specifically, a regularized (16) is solved in each iteration

$$\mathbf{h}^{(l,s)} = \arg\max_{\mathbf{h}\in\mathbb{R}^L} \frac{\mathbf{h}^T \left[\alpha \mathbf{C}_1^w - (1-\alpha)\,\Phi^{(l,s-1)}\right]\mathbf{h}}{\mathbf{h}^T \mathbf{C}_\Sigma^w \mathbf{h}}, \tag{26}$$

where

$$\mathbf{C}_\Sigma^w = \mathbf{C}_1^w + \mathbf{C}_2^w. \tag{27}$$

Diagonal matrix $\Phi^{(l,s-1)}$ is defined similar to $\mathbf{\Omega}^{(l,s-1)}$ by replacing $w_i^{(l,s-1)}$ in (25) by $h_i^{(l,s-1)}$. The major steps of the iterative reweighting method are summarized in Algorithm 2.

Algorithm 2 Iterative Reweighting Method

1: Set $s = 0$ and initialize diagonal elements of $\mathbf{\Omega}^{(l,0)}$ (or $\Phi^{(l,0)}$) as $1/N$ (or $1/L$);
2: **while** not converged **do**
3: Set $s = s + 1$;
4: Update $\mathbf{\Omega}^{(l,s-1)}$ (or $\Phi^{(l,s-1)}$);
5: Solve (24) [or (26)] for $\mathbf{w}^{(l,s)}$ (or $\mathbf{h}^{(l,s)}$);
6: **end while**
7: Return the final solution as $\mathbf{w}^{(l)}$ (or $\mathbf{h}^{(l)}$) to Algorithm 1.

It is worth noting that, without the sparsity regularization, Algorithm 1 based on the alternating optimization strategy is guaranteed to converge, although the final results are not surely the globally optimal solutions to the original problem. In contrast, so far there is no convergence guarantee of the iterative reweighting method presented in Algorithm 2.

TABLE I
PERFORMANCE OF FOUR ALGORITHMS USING DATA-I

	CSP	JSTFD	STRCSP	TSLDA
SVM	81.92	**85.56**	83.64	–
LDA	83.72	**85.77**	84.51	82.27
mean	82.82	**85.67**	84.08	82.27

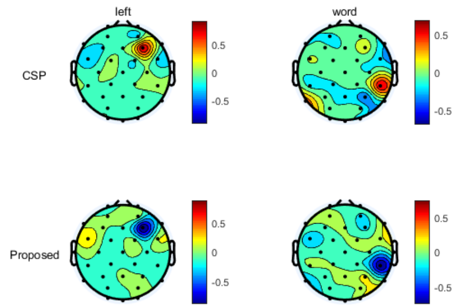


Fig. 1. Scalp weight topography of spatial filters obtained by traditional CSP algorithm and the proposed JSTFD algorithm for Data-I.

Thus, outer and inner iterative procedures could be terminated when iteration numbers reach T_1 and T_2, respectively. But experimental results show that the proposed algorithm can empirically converge to final solutions in a few iterations. This will be verified by convergence analyses based on experimental results presented in Subsection IV-E.

IV. EXPERIMENTAL RESULTS

A. Experimental Setup

The data used in our experiments are taken from Dataset V of BCI Competition III [29] and Dataset IIa of BCI Competition IV [30]. Dataset V of BCI Competition III are collected from three subjects with sampling rate 512 Hz, which includes 3 MI tasks (i.e., left hand, right hand, and word associated with labels 2, 3, and 7, respectively). But only two of them (i.e., tasks with labels 2 and 7) are employed in our evaluation. Dataset IIa of BCI Competition IV involves 4 MI tasks (i.e., left hand, right hand, both feet, and tongue) from 9 healthy subjects. We only use the data associated with left- and right-hand movements. EEG signals are recorded through 22 channels with sampling frequency 250 Hz. EEG signals of both datasets are filtered by a bandpass filter with passband from 9 to 35 Hz before feature extraction. For the convenience of reference, these two datasets are denoted in the subsequent discussion by "Data-I" and "Data-II", respectively.

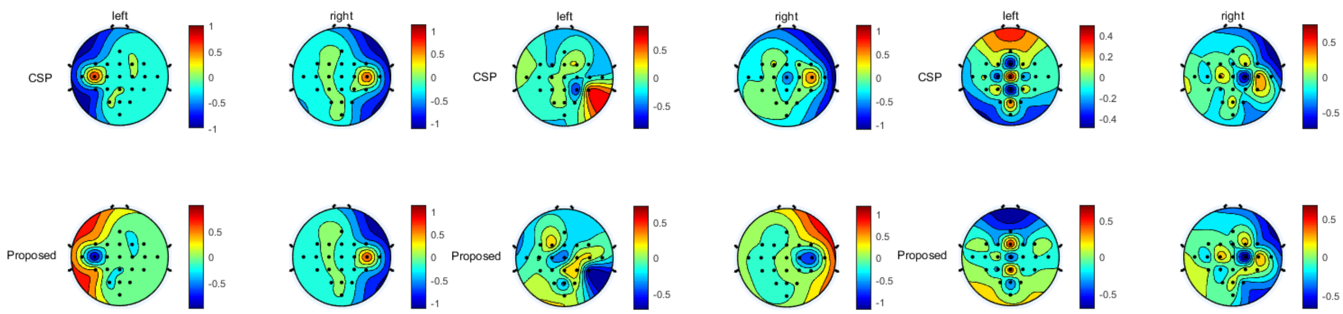
Both LDA and SVM with Gaussian kernel function are adopted as classifiers in our experiments. For CSP-based approaches, a classifier is implemented on the logarithm of signal variances after filtering. To enhance the generalization capability, 10 × 10-fold cross validation is employed in our experiments. Average classification accuracies are presented as experimental results. All the experiments are performed by MATLAB 2018a running on Linux operating system with Intel(R) Xeon(R) CPU E5-2620 v4 @ 2.10GHz processor and RAM 64 GB.

B. Performance Comparison of Existing Algorithms

We compare classification accuracies of the proposed JSTFD algorithm with those of the other three state-of-the-arts

TABLE II
PERFORMANCE OF FOUR ALGORITHMS USING DATA-II

Algorithm	Subjects									mean±std
	S01	S02	S03	S04	S05	S06	S07	S08	S09	
CSP+SVM	83.2	53.7	93.8	71.6	51.8	63.3	71.9	94.8	80.4	73.8 ± 15.7
CSP+LDA	**86.6**	54.9	94.4	**76.1**	56.8	**69.4**	76.4	95.2	83.7	77.1 ± 14.7
JSTFD+SVM	84.1	53.1	94.6	71.0	88.5	56.0	73.0	94.1	84.6	77.7 ± 15.4
JSTFD+LDA	86.4	**55.9**	**96.3**	73.1	**89.5**	58.2	76.1	93.8	**86.6**	$\mathbf{79.6 \pm 14.7}$
STRCSP+SVM	83.4	50.9	92.7	57.1	88.1	58.8	66.9	95.4	70.2	73.7 ± 16.6
STRCSP+LDA	85.5	50.8	95.6	62.2	88.9	62.2	73.5	**95.5**	69.5	76.0 ± 16.1
TSLDA	80.5	51.3	87.5	59.3	45.0	55.3	**82.1**	84.8	86.1	70.2 ± 17.1



(a) Subject 3 (b) Subject 5 (c) Subject 9

Fig. 2. Scalp weight topography of spatial filters obtained by traditional CSP algorithm and the proposed JSTFD algorithm for Subjects 3, 5, and 9 in Data-II.

methods, i.e., traditional CSP, spatio-temporally regularized CSP (STRCSP) [14], and tangent space LDA (TSLDA) [21]. As its name implies, the TSLDA method is equipped with the LDA classifier. Traditional CSP, STRCSP, and the proposed algorithm learn features from data samples, that are fed to either SVM or LDA. In the experiments of this subsection, we only use one pair of spatial filters in the proposed JSTFD algorithm and the other two CSP-based approaches.

For Data-I, we choose $\alpha = 0.1$ and $L = 40$. Table I lists classification accuracies of all the four algorithms. The mean classification accuracy of each algorithm is also given in Table I for comparison. The best results are highlighted in bold font. It can be observed that, compared to the other three approaches, our algorithm can achieve 2.85%, 1.59%, and 3.40% performance improvement, respectively.

For Data-II, α and L are set, respectively, to 0.1 and 20. Classification accuracies of each algorithm are summarized in Table II. The highest classification accuracy is also marked in bold font for each subject. Among all the nine sets of experiments, the proposed JSTFD algorithm equipped with LDA achieves the best results in four cases, while traditional CSP with LDA outperforms the other approaches in another three cases. To compare the average performance of each algorithm, the mean and standard deviation of classification accuracies achieved by each algorithm are also listed in Table II. Compared to traditional CSP, STRCSP, and TSLDA, the proposed algorithm achieves at least 2.5% performance improvement.

Figs. 1 and 2 show scalp weight topographies of spatial filters obtained by traditional CSP and the proposed JSTFD algorithm for Data-I and also EEG data of Subjects 3, 5, 9 in Data-II, respectively. It can be observed that spatial weights obtained by these two methods demonstrate similar patterns.

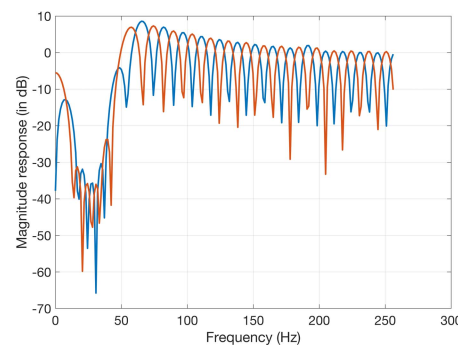
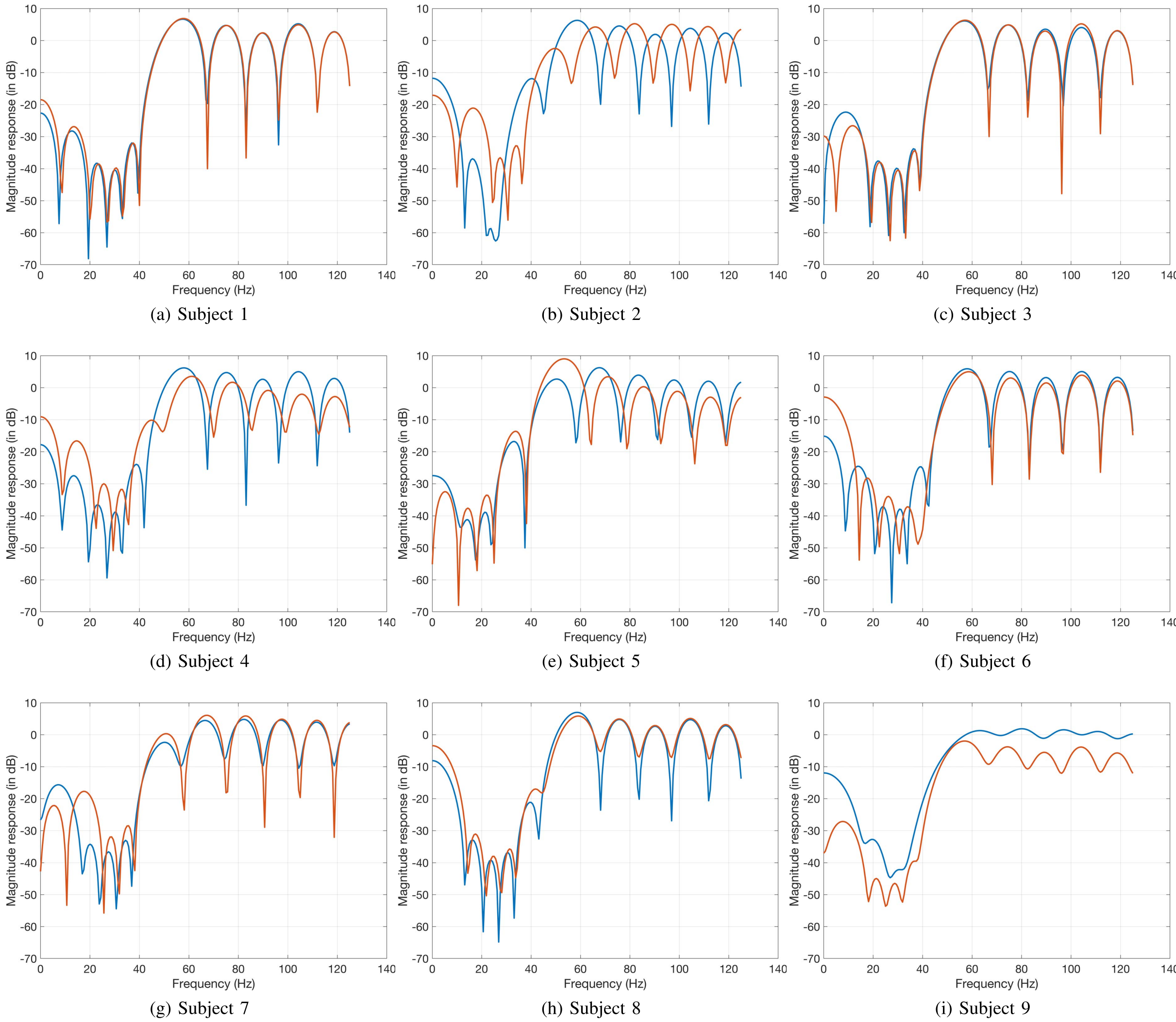


Fig. 3. Magnitude responses of two temporal filters for Data-I.

Temporal filters are introduced in the proposed algorithm to improve classification performance. They are trained by EEG data to maximize the ratio of covariances of jointly filtered EEG data in two classes. Fig. 3 shows magnitude responses of two temporal filters obtained in the experiment of Data-I. Both filters suppress low frequency components below about 50 Hz. Beyond this critical frequency point, they both work like comb filters, which have a series of peaks and notches in their frequency responses. However, frequency points corresponding to magnitude peaks of one temporal filter are just located at those corresponding to magnitude notches of the other filter, yielding an orthogonal decomposition in frequency domain. Magnitude responses of temporal filters designed for Data-II are depicted in Fig. 4. They demonstrate similar spectral properties as in Fig. 3. But feature extraction in motor imagery tasks is generally subject-dependent. This is also reflected by frequency responses of temporal filters designed for EEG data of each subject. For instance, Fig. 4a shows that magnitude peaks of two filters coincide with each other, which



(a) Subject 1 (b) Subject 2 (c) Subject 3

(d) Subject 4 (e) Subject 5 (f) Subject 6

(g) Subject 7 (h) Subject 8 (i) Subject 9

Fig. 4. Magnitude responses of two temporal filters for Data-II.

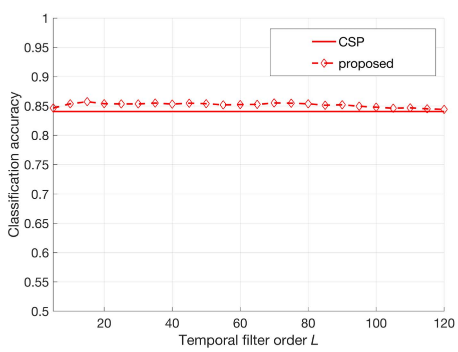


Fig. 5. Variation of classification accuracies with respect to L using Data-I.

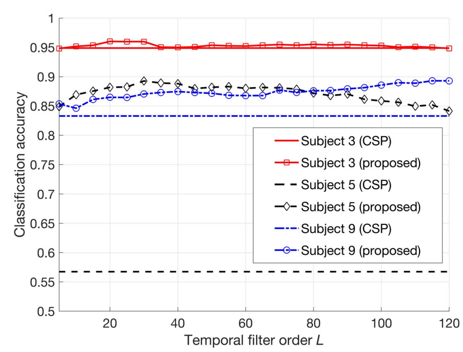
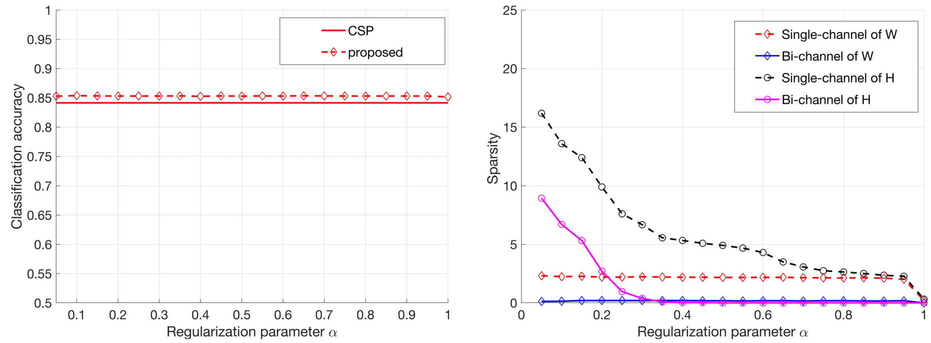


Fig. 6. Variation of classification accuracies with respect to L using datasets of Subjects 3, 5 and 9 in Data-II.

is different to Fig. 4b, even though Subjects 1 and 2 perform the same motor imagery task.

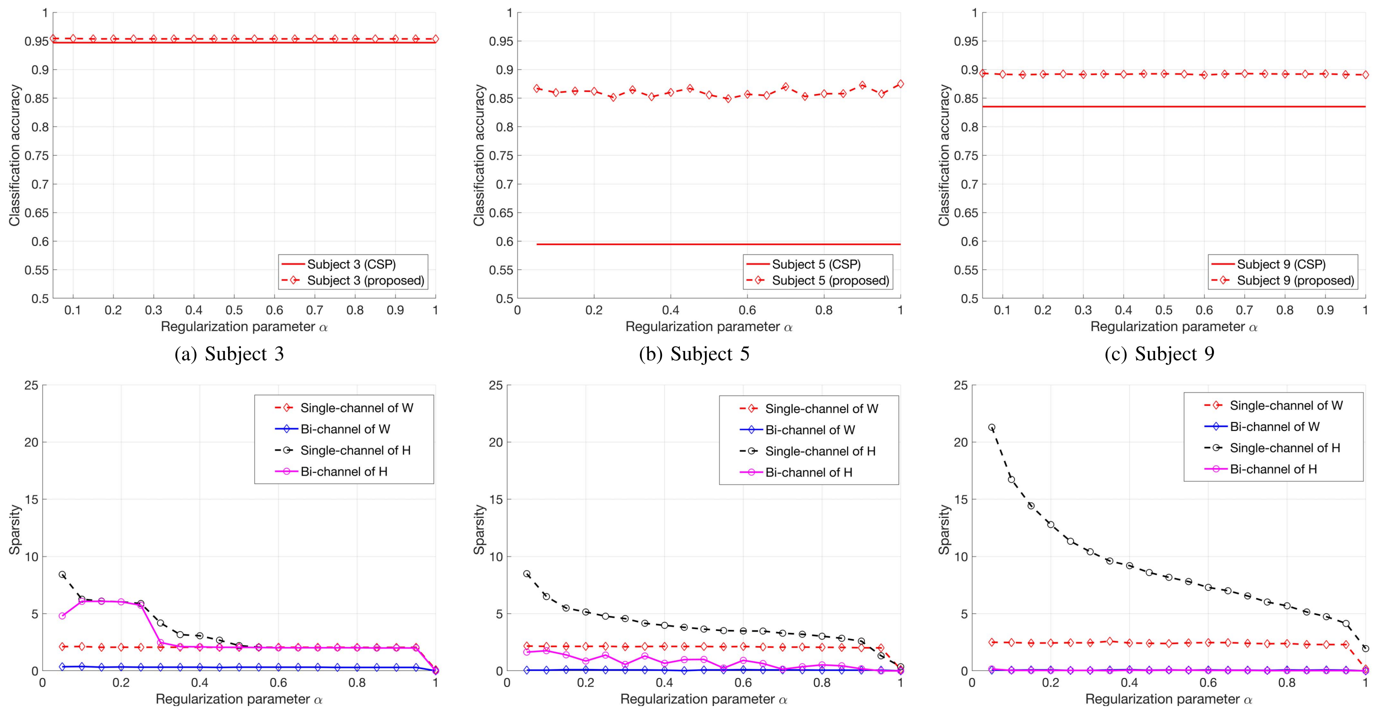
C. Effects of L and α

In the proposed JSTFD algorithm, temporal filter order L and regularization parameter α need to be chosen appropriately. To evaluate their impacts on final classification accuracies, we further conduct a set of experiments by fixing one parameter and varying the other ones. Experimental results of the proposed algorithm are compared with those of traditional CSP. Both feature extraction algorithms are equipped with the LDA classifier. In the evaluation of L and α, we use only one spatial filter for each class of EEG signals, that is, $N = 1$. Data-I and also EEG data of Subjects 3, 5 and 9 in Data-II are employed in our experiments.



(a) (b)

Fig. 7. Variation of classification accuracies and sparsities of filter coefficients with respect to α using Data-I.



(a) Subject 3 (b) Subject 5 (c) Subject 9

(d) Subject 3 (e) Subject 5 (f) Subject 9

Fig. 8. Variation of classification accuracies and sparsities of filter coefficients with respect to α using Data-II.

First of all, temporal filter order L varies from 5 to 120 with incremental step equal to 5, while α is fixed equal to 0.1. Figs. 5 and 6 depict the variation of classification accuracies with respect to L for Data-I and Data-II, respectively. Experimental results of traditional CSP are also given in Figs. 5 and 6 as baselines. It can be observed that the proposed JSTFD algorithm achieves better performance than CSP when L is appropriately chosen. For Data-I, the best L is around 15, yielding an 1.5% improvement on classification accuracy. Data-II contains datasets of different subjects. Since feature extraction is generally subject-dependent, different levels of performance improvement are achieved in experiments of different subjects. For example, the proposed algorithm exploiting temporal filters can improve classification accuracy by at least 28% in experiments of Subject 5, while at most 2% improvement is achieved for Subject 3. Temporal filters with a smaller order cannot sufficiently exploit short-term correlation

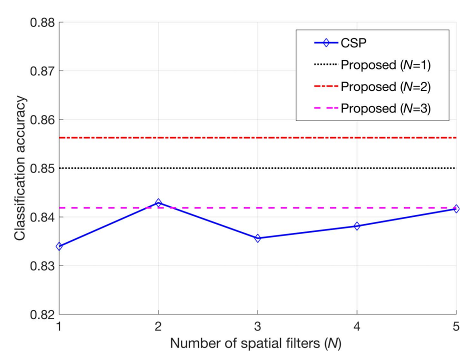
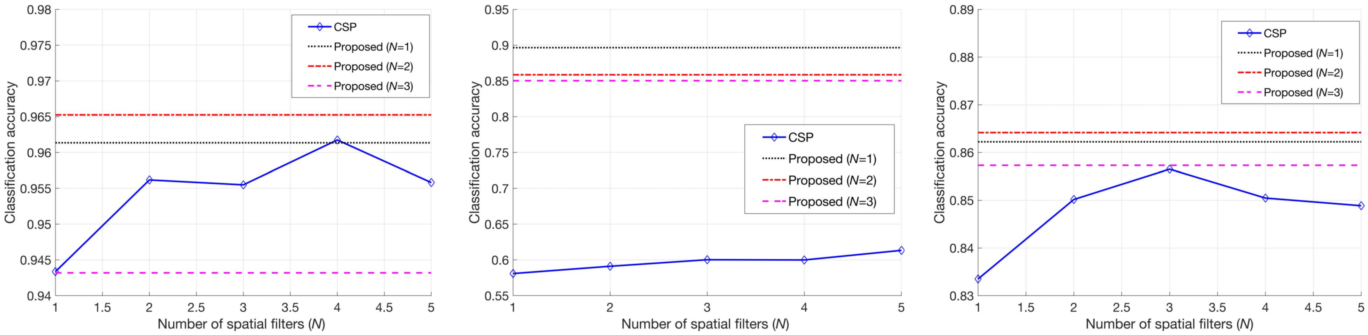


Fig. 9. Variation of classification accuracies with respect to N using Data-I.

of EEG signals in time domain, whereas temporal filters with a larger order cannot cope with the essential nonstationarity of EEG signals. Thus, with the increase of L, the performance



(a) Subject 3 (b) Subject 5 (c) Subject 9

Fig. 10. Variation of classification accuracies with respect to N using Data-II. (a) Subject 3. (b) Subject 5. (c) Subject 9.

of the JSTFD algorithm is generally improved first and then deteriorates. But the best L is usually not the same (e.g., 20, 30, 115 for Subjects 3, 5 and 9 in Data-II). So in order to achieve the highest classification accuracy, L has to be tuned in the training stage. In practice, if K is sufficiently large, L between 20 and 60 is generally sufficient for the proposed algorithm to achieve good classification performance.

Parameter α is used to control the sparsity of both spatial and temporal filters. To evaluate its effects on classification accuracies, we further vary α from 0.05 to 1 with incremental step 0.05, while temporal filter order L is chosen as 15 for Data-I and 20, 30, 115 for the experiments of Subjects 3, 5 and 9 in Data-II. The variations of classification accuracies and sparsities of both spatial and temporal filter coefficients with respect to α are shown in Figs. 7 and 8. Here, the sparsity of a filter is computed by the average number of zeros existing in $\mathbf{w}$ or $\mathbf{h}$. For binary classification, there are at least two pairs of spatial and temporal filters. If zeros appear at the same positions of a pair of filters, then the corresponding EEG channel or arithmetic computation can be abandoned. In view of this perspective, besides "single-channel" sparsities which count the total number of zero coefficients in all the spatial or temporal filters, we also present "bi-channel" sparsities in Figs. 7 and 8, which count the number of zero coefficients existing at the same positions of a pair of spatial or temporal filters. It can be observed that the regularization term of sparsity has a very limited impact on sparsities of spatial filters. Even for different subjects, the proposed algorithm achieves spatial filters with almost the same sparsity level. In contrast, with the decrease of α, more temporal filter coefficients could be nullified. Experimental results also reveal that the fluctuation of classification accuracies with respect to α is very flat, which indicates that more zero coefficients do not always lead to performance degradation. Thereby, a small α (e.g., $\alpha = 0.1$) is preferred in practice.

D. Effects of N

Although the order of spatial filter is determined by the number of EEG channels, in general one can improve the classification accuracy by employing more spatial filters. But a large N also increases the overfitting risk, possibly yielding severe performance degradation. To evaluate its effects, in this experiment we first fix α equal to 0.1 and L is set, respectively,

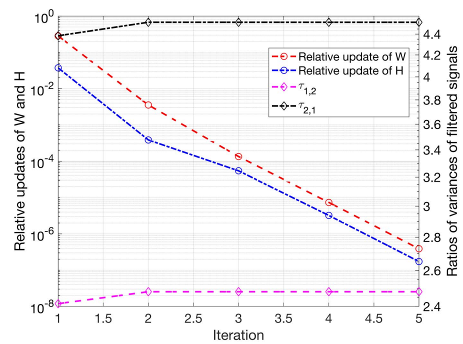
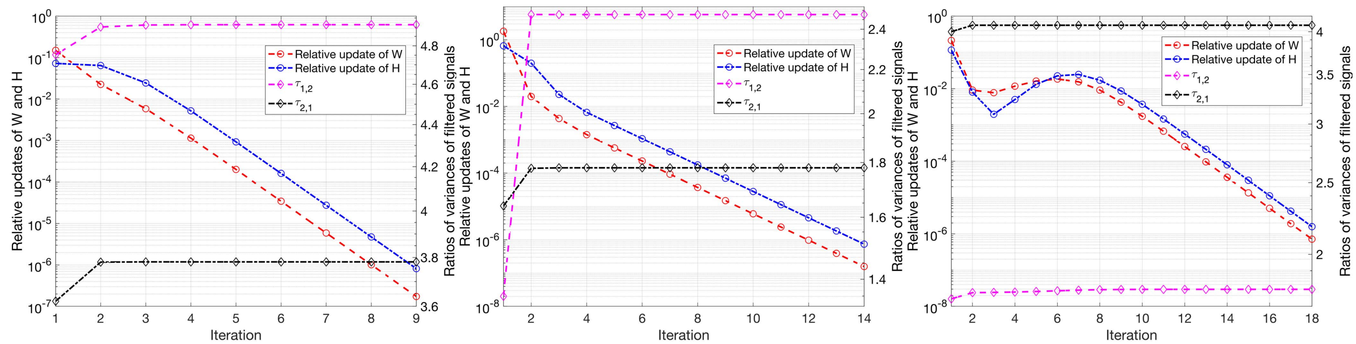


Fig. 11. Variations of $\Delta\mathbf{W}^{(l)}$, $\Delta\mathbf{H}^{(l)}$ and $\tau_{1,2}$, $\tau_{2,1}$ with respect to iteration number l in experiment using Data-I.

to 15 for Data-I and 20, 30, 115 for Subjects 3, 5 and 9 in Data-II. For traditional CSP, N varies from 1 to 5, whereas we choose N between 1 and 3 for the proposed JSTFD algorithm. Classification features extracted by both algorithms are all fed to the LDA classifier.

Fig. 9 illustrates the variation of classification accuracies with respect to N for Data-I. Obviously, traditional CSP and the proposed JSTFD algorithm both achieve the best results when $N = 2$. But the proposed algorithm outperforms traditional CSP by about 1.5% in terms of classification accuracy. It is also noticed that, compared to traditional CSP using multiple spatial filters, feature extraction based on joint spatio-temporal filtering is a more efficient way to improve classification performance. In this set of experiments, the proposed JSTFD algorithm using two pairs of spatial filters and one pair of temporal filters outperforms traditional CSP even using five pairs of spatial filters.

Fig. 10 shows experimental results of Subjects 3, 5 and 9 in Data-II. Similarly, N is chosen between 1 and 3, while classification performance of traditional CSP is evaluated using $N = 1 \sim 5$. As in the previous experiments of Data-I, the proposed algorithm achieves the best results by $N = 2$ for Subjects 3 and 9. In contrast, more spatial filters are required by traditional CSP to obtain the peak scores of classification accuracies. For Subject 5, more than 25% improvement in terms of classification accuracy is achieved by joint spatio-temporal filtering, compared to traditional CSP. Furthermore, using more than one pair of spatial filters leads to performance degradation.



(a) Subject 3 (b) Subject 5 (c) Subject 9

Fig. 12. Variations of $\Delta\mathbf{W}^{(l)}$, $\Delta\mathbf{H}^{(l)}$ and $\tau_{1,2}$, $\tau_{2,1}$ with respect to iteration number l in experiment using Data-II. (a) Subject 3; (b) Subject 5; (c) Subject 9.

TABLE III
AVERAGE CPU TIME OF DIFFERENT ALGORITHMS

Algorithm	CPU Time (sec)			
	Data I	Data-II		
		Subject 3	Subject 5	Subject 9
CSP+LDA	0.07	0.03	0.03	0.02
JSTFD+LDA	10.86	17.78	23.39	17.11
STRCSP+LDA	6.56	3.39	3.15	2.96
TSLDA	12.92	13.03	12.62	11.8

E. Convergence Analyses & Computational Complexity

In this subsection, we empirically evaluate the convergence and computational complexity of the proposed algorithm using Data-I and Data-II of Subjects 3, 5, and 9. In our experiments, α is always set to 1, while L is chosen as 40 and 20 for Data-I and Data-II, respectively.

The convergence analysis of the proposed algorithm is presented first. We adopt here the relative updates of spatial or temporal filters, i.e., $\Delta\mathbf{W}^{(l)} = \frac{\|\mathbf{W}^{(l)}-\mathbf{W}^{(l-1)}\|_F}{\|\mathbf{W}^{(l-1)}\|_F}$ and $\Delta\mathbf{H}^{(l)} = \frac{\|\mathbf{H}^{(l)}-\mathbf{H}^{(l-1)}\|_F}{\|\mathbf{H}^{(l-1)}\|_F}$, and also ratios of variances of filtered EEG signals in different classes, i.e., $\tau_{1,2} = \frac{\mathbf{w}^T\mathbf{C}_1^h\mathbf{w}}{\mathbf{w}^T\mathbf{C}_2^h\mathbf{w}}$ or $\tau_{2,1} = \frac{\mathbf{w}^T\mathbf{C}_2^h\mathbf{w}}{\mathbf{w}^T\mathbf{C}_1^h\mathbf{w}}$. Figs. 11 and 12 illustrate the variation of $\Delta\mathbf{W}^{(l)}$, $\Delta\mathbf{H}^{(l)}$, $\tau_{1,2}$, and $\tau_{2,1}$ with respect to iteration number l for Data-I and Data-II of Subjects 3, 5, and 9, respectively. It can be observed that the proposed algorithm can converge in a fast speed.

To evaluate computational efficiency of each algorithm, we list the average CPU time in Table III. As noticed, the proposed algorithm consumes more computation compared to the other approaches, due to its iterative essence. But computational costs per iteration are still low, since optimization problems (12) and (16) [also, (24) and (26)] are efficiently solved by the generalized EVD.

V. CONCLUSIONS

A novel spatio-temporal filter design algorithm has been presented for binary classification of MI BCI systems. To improve computational efficiency and alleviate overfitting, a common temporal filter is shared by each channel of EEG signals in the same class. The proposed JSTFD algorithm adopts an alternating optimization scheme to update spatial and temporal filters, such that the resulting problems can still be cast in a CSP form and, thus, efficiently solved by the EVD. To improve the sparsity, a regularization term is introduced in the spatial and temporal filter designs and an iterative reweighting technique is employed to tackle the resulting problems. Experimental results using public datasets of BCI Competition III and and IV have demonstrated the effectiveness of the proposed algorithm.

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